RoverSetter

Three physical prototypes. One stability problem. A different engineering approach.

01 The problem originated in my business

An installation through Villegas Headstones & Monuments took approximately three to four hours, and the stone was damaged during handling. I replaced it at my expense. I began RoverSetter to investigate whether a compact mobile machine could simplify monument movement and precise placement.

THREE PHYSICAL PROTOTYPE ITERATIONS

Prototype 1

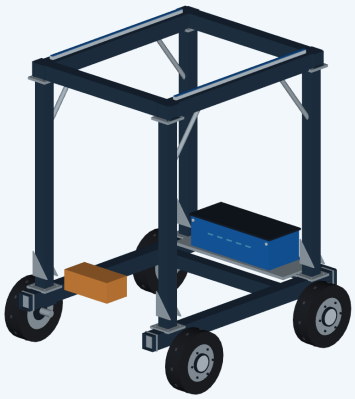
Built an initial mechanical proof-of-concept; exact early records unavailable.

Prototype 2

Revised mechanisms after I was dissatisfied with the earlier design; changes not fully recovered.

Prototype 3

Integrated motorized motion with a forward hoist/positioning concept; tested a smaller stone.



16 in across × 8 in front-back × 6 in high

02 What happened in Prototype 3

In the third prototype test, I used a roughly 70-pound rectangular stone suspended beyond the front wheel line on a hard surface. Only the wheels supported the chassis. While stationary, both rear wheels lifted nearly together and the machine rocked back. Forward pitching became more pronounced immediately when the drive motors started accelerating the chassis.

IMPORTANT DISTINCTION

This is my retrospective recollection. The stone location, hoist geometry, original force measurements and exact movement history have not been independently recovered.

What I overlooked - and what I could actually test

03 The design-assumption finding

In my original stability calculations, I treated the stone as remaining at one fixed position. But I had designed a mechanism that could move it forward relative to the wheel supports. I did not evaluate stability throughout that intended travel. This is my recollection of the original analysis; the old calculation sheets have not been recovered.

STATIC MODEL / NOT HISTORICAL TEST DATA

$$N_{rear} = (M g L - m g d) / B$$

POSITIONING MECHANISM

Moving the stone farther ahead of the front wheels increases the gravitational overturning moment. In the simplified level-ground static model, the predicted total rear reaction decreases.

DRIVETRAIN STARTUP

Forward pitching increased when the motor-powered chassis accelerated. Forward acceleration alone does not prove this effect; transient forces, swing, contact and structural motion remain unresolved.

04 What the investigation established

The 2026 numerical model reproduced the v11 hypothetical scenarios and quantified sensitivity to forward load overhang. It did not reproduce the original failure, measure wheel reactions, or validate a redesigned machine. The saved CAD files preserve the original design dimensions, according to my design records and recollection; they are not a substitute for measured wheel forces or historical test data.

DOCUMENTED / REPRODUCIBLE

3 owner-reported physical builds; 16 × 8 × 6-in stone recollection; owner-checked design dimensions; reproducible static equations and code.

NOT YET DEMONSTRATED

Original tipping root cause; experimental wheel-force data; teacher review; independently validated full-scale redesign or lifting capacity.

05 Next step: independent evidence

A physics or engineering teacher could approve a harmless stationary tabletop analogy, using a very small attached mass and measuring how rear support changes with position. The experiment would provide new classroom-scale data - not proof that the original heavy-load machine was safe.

CENTRAL LESSON / A mechanism can work as intended while the complete system fails to remain stable throughout its operating conditions.

Design file note: these CAD views come from updated exports of my existing RoverSetter models. Original prototype photographs and measured test forces are unavailable.